# Skander Moalla

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## EDUCATION

**EPFL** **Lausanne, Switzerland**
*PhD in Computer and Communication Sciences* *Sep 2022 – Present (graduation expected in Fall 2026)*
Learning from Experience: Plasticity, Reinforcement, and Exploration. Advised by Prof. Caglar Gulcehre.
Deep reinforcement learning, LLM post-training, diversity and test-time scaling, agentic formal math proving.

**University of Oxford** **Oxford, UK**
*MSc in Advanced Computer Science* *Oct 2021 – Sep 2022*
MSc thesis in multi-agent reinforcement learning with Prof. Shimon Whiteson's group (WhiRL).

**École Polytechnique** **Paris, France**
*BSc in Mathematics and Computer Science (Double Major)* *Sep 2017 – Aug 2020*
Graduated Summa Cum Laude — CGPA: 4.09/4.0 — Awarded an excellence scholarship.
Exchange programs: University of Toronto (Fall 2019) & Stanford University (Summer 2019).

## EXPERIENCE

**Meta FAIR** **Paris, France**
*Research Scientist Intern* *Mar 2026 – Aug 2026*
- Hosted by Julia Kempe and Rémi Munos in the Foundations of Reasoning / Math Superintelligence Team.
- Working on agentic in-context adaptation, self-improvement, and asymmetric self-play in formal math (Lean).

**Google DeepMind** **Paris, France**
*PhD Student Researcher* *Sep 2025 – Dec 2025*
- Hosted by Alexandre Ramé in the Gemma post-training team (Olivier Bachem, Lucas Dixon).
- Worked on leveraging diversity to improve test-time scaling (pass@k) and sampling in reinforcement learning.
- Contributed to the Gemma open-source JAX codebase (Gemma 3 KV cache sharing and sampler).

## PUBLICATIONS

Simon Matrenok*, **Skander Moalla***, and Caglar Gulcehre. Quantile Reward Policy Optimization: Alignment with Pointwise Regression and Exact Partition Functions. ***NeurIPS*, 2025**.

Project Apertus, [...], **Skander Moalla***, [...], Antoine Bosselut, Martin Jaggi, and Imanol Schlag. Apertus: Democratizing Open and Compliant LLMs for Global Language Environments. ***ACL*, 2026**.

**Skander Moalla**, Andrea Miele, Razvan Pascanu, and Caglar Gulcehre. No Representation, No Trust: Connecting Representation, Collapse, and Trust Issues in PPO. ***NeurIPS*, 2024**.

Xiuying Wei, **Skander Moalla**, Razvan Pascanu, and Caglar Gulcehre. Building on Efficient Foundations: Effectively Training LLMs with Structured Feedforward Layers. ***NeurIPS*, 2024**.

**Skander Moalla***, Manuel Madeira*, Lorenzo Riccio*, and Joonhyung Lee*. [Re] Reproducibility Study of Behavior Transformers. ***ReScience C*, 9(2), 2023. Outstanding Paper Awards Honorable Mentions**.

Benjamin Ellis, Jonathan Cook, **Skander Moalla**, Mikayel Samvelyan, Mingfei Sun, Anuj Mahajan, Jakob Foerster, and Shimon Whiteson. SMACv2: An Improved Benchmark for Cooperative Multi-Agent Reinforcement Learning. ***NeurIPS*, 2023**.

## EXPERIENCE (CONTINUED)

**Quincus** — **Singapore (Remote)**
*Applied Research Intern* — *Mar 2021 – Aug 2021*

- Deployed a solution leveraging deep reinforcement learning (RL) to optimize middle-mile logistics.
- Built simulators to model multi-mile parcel shipping and trained deep RL prototypes (PPO, DQN).
- Stack: Python (OpenAI Gym & Baselines, LightGBM, pytest), Docker, NSCC High-Performance Computing.

**Amazon** — **Luxembourg**
*Software Development Intern* — *Sep 2020 – Feb 2021*

- Completed full-stack and DevOps sprint tasks within a "two-pizza team" owning multiple services end-to-end.
- Designed and built a prototype to detect similar and duplicate business metrics supporting Amazon's transportation network worldwide to ease metric discovery and reduce computation costs leveraging deep learning and NLP techniques with AWS SageMaker.
- Stack: AWS, Python, Java (Dagger, AutoValue, JUnit, Mockito), TypeScript (ImmutableJS, Mocha, Chai).

**EPFL – Laboratory for Topology and Neuroscience & Blue Brain Project** — **Geneva, Switzerland**
*Research Intern* — *Jan 2020 – Apr 2020*

- BSc thesis supervised by Prof. Kathryn Hess Bellwald (Link to thesis).
- Built a framework for neuron simulation and leveraged tools from information theory and algebraic topology (simplicial homology) to formulate new conclusions about the dynamics of neurons forming directed cliques.
- Stack: Python 3 (DIT, JupyterLab, NumPy, SciPy, Brian2), C++.

**Ooredoo** — **Tunis, Tunisia**
*Full-Stack Developer Intern* — *Jul 2018 – Aug 2018*

- Implemented a PoC for a custom notification system in a hybrid mobile app featuring profiling functionalities and allowing Ooredoo to adjust the activity of their notification server. Stack: Ionic 3, Node.js, socket.io.

## SOFTWARE

### Core stack and responsibilities

**Co-lead of the Swiss AI Apertus v1 post-training** — *July 2024 – Present*

- Developed infrastructure and offline post-training code to train the Swiss open models Apertus 8B and 70B on the CSCS Alps cluster with 1000+ GH200 nodes.
- Supervised MS students on chat templates, evaluations, supervised fine-tuning, multilinguality, and data.

**Point of Contact for infrastructure & compute platforms in the lab** — *Sep 2023 – Present*

- Support the deployment of container-based research experiments on multiple clusters at the scale of hundreds of nodes, including Slurm HPC clusters and Run:AI (Kubernetes-based) clusters. (Documentation)
- Stack: Docker, Kubernetes, Run:AI, Slurm, NVIDIA Pyxis, NVIDIA Enroot, Apptainer/Singularity.

### Open-source contributions and selected projects

**Open-source contributions**

- @PyTorch/RL, @DLR-RM/stable-baselines3, @Farama-Foundation/Gymnasium, @HuggingFace/TRL.

**Python Machine Learning Research Project Template** — *Sep 2022 – Present*

- Maintain a template for reproducible Python machine learning research projects deployable on various platforms including Slurm (with containers through NVIDIA Pyxis/Enroot or Apptainer) and Kubernetes clusters.

## TEACHING

**Université Dauphine Tunis (Université Paris Dauphine-PSL)** — **Tunis, Tunisia**
*Visiting Lecturer – Reinforcement Learning* — *Jan 2026*

- Delivered a 24-hour course for the Master's in Big Data & Artificial Intelligence (BDIA).
- Covered reinforcement learning foundations and modern LLM post-training methods such as RLHF.

**EPFL** **Lausanne, Switzerland**
*LauzHack Deep Learning Bootcamp Instructor (Deep Reinforcement Learning)* *Jul 2024*

- Delivered a 2-hour introductory lecture on deep reinforcement learning (Bellman equations, DQN, PPO, etc.).

**EPFL** **Lausanne, Switzerland**
*Teaching Assistant & Supervisor* *Sep 2022 – Present*

- Supervised 9 Master's projects and theses.
- Probability and Statistics (MATH-232 Spring 2023), Machine Learning (CS-433 Fall 2023 – exam preparation), Artificial Neural Networks / Reinforcement Learning (CS-456 Spring 2024 – project preparation).

**ATSM – Tunisian Association of Mathematics** **Tunis, Tunisia**
*Olympiad Coach* *Jul 2018 – Present*

- Designed material for Olympiad problem-solving and trained national team members for the Mediterranean Mathematical Olympiad (MYMC 2018) and the International Mathematical Olympiad (IMO 2020).

## SOFTWARE - CLASS PROJECTS

**SAM - Study Abroad Matching** *Sep 2019 – Dec 2019*

- Designed a web app around an algorithm to match students applying to study-abroad programs and universities based on a human-in-the-loop solution to the Gale–Shapley college admissions problem.
- Stack: Python, Django, React, Google Cloud Platform (GCP), Google App Engine, Google Cloud SQL.

**Computer Facial Discrimination - Machine Learning Project** *May 2019 – Jun 2019*

- Studied the effect of different pre-trained feature extractors (Autoencoders, CNN classifiers) and clustering methods on the discrimination of human faces.
- Built a customizable autoencoder model to demonstrate that training data (as an analogy for background and culture) can be the root of multiple kinds of discrimination.
- Stack: Python, TensorFlow, Keras, scikit-learn, Google Colab, Kaggle.

**TOP5 Basketball Team Manager - C++ Project** *Nov 2018 – Jan 2019*

- Managed a team of 9 students / Led the Game Engine team / Maintained the Git repository.
- Designed and implemented the game engine and its interaction with the remaining game entities.
- Stack: C++, Qt.

**Meet Halfway - Web Project** *May 2018 – Jun 2018*

- Built a user-friendly website that allows a group of friends to find the best destination to travel to and meet for holidays according to their respective locations and cost of travel.
- Stack: HTML, CSS, JS.

## EDUCATION - EXCHANGE PROGRAMS

**University of Toronto** **Toronto, Canada**
*Exchange – Fall 2019 — Software Engineering, Databases, Compilers & Interpreters* *Sep 2019 – Dec 2019*

**Stanford University** **Stanford, USA**
*Exchange – Summer 2019 — GPA: 4.15/4.0 — AI, Entrepreneurship & Innovation* *Jun 2019 – Aug 2019*

## LANGUAGES

English: Bilingual French: Native Arabic: Native German: Elementary